

MINING HOTEL BOOKINGS TO PREDICT CANCELLATIONS

SEIJI AOYAMA &
ELAINE FRENCH



PROBLEM STATEMENT

- Hotels struggle with maintaining stable occupancy due to unexpected cancellations.
- This analysis uses machine learning to predict the likelihood of booking cancellations.
- Goal: Provide data-driven insights that help hotels anticipate cancellations early on.

QUESTIONS SOUGHT TO ANSWER

1. Can we accurately predict which reservations are likely to be canceled, well before the arrival date?
2. Which factors have the strongest influence on cancellation risk?

DATASET

We are using the hotel booking demand dataset found at

<https://www.kaggle.com/datasets/mojtaba142/hotel-booking/data>

- This data was provided by **two hotels in Portugal: a resort hotel in the Algarve region (H1) and a city hotel in Lisbon (H2).**
- The authors of the paper, Nuno Antonio, Ana de Almeida, and Luis Nunes collected and published the data with the cooperation of the hotels.

DATA PREPARATION WORK

Data clean/preprocess:

- Verified completeness and quality of key variables (e.g., is_canceled, lead_time) and removed irrelevant identifiers.
- Applied encoding for categorical variables and performed feature engineering (e.g., total_stay, lead_time_log) guided by EDA insights.
- Ensured a robust, well-structured dataset by addressing skewness, outliers, and potential multicollinearity.

TOOLS USED

Language

- Python



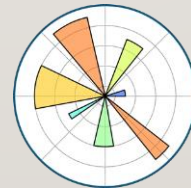
Library

- Pandas
- Numpy
- Sklearn
- Matplotlib
- Scipy
- Seaborn



Others

- Github
- Jupyter notebook



MODELING APPLIED

Data-driven Models:

- Decision Tree
- Logistic Regression
- Naïve Bayes

Performance:

- Decision Tree: Best performer with ~85% accuracy and high recall (81%).
- Logistic Regression: Solid baseline (78.8% accuracy) but lower recall.
- Naïve Bayes: Simple, moderate performance (~76% accuracy).

Key Predictors:

- Lead time, deposit type, and average daily rate significantly influenced cancellation likelihood.
- Implications: Insights support strategies like adjusting deposit policies, pricing, and targeted interventions to reduce cancellations.

VALIDATION APPLIED

1. 5-Fold Cross-Validation

- Reduce bias from a single data split and confirms model stability and consistency.

2. ROC-AUC

- Examined ROC curves and AUC scores to compare model trade-offs between true positive and false positive rates.
- Logistic Regression showed a balanced performance
- Decision Tree's curve indicated strong early true positive detection with fewer false positives, demonstrating its overall robustness.

KNOWLEDGE GAINED

1. Data Preprocessing:

Learned to handle missing values, encode categorical data, and address skewness for clean, analysis-ready datasets.

2. Feature Engineering:

Developed domain-specific features and applied EDA to identify key patterns and refine predictors.

3. Model Building & Evaluation:

Gained hands-on experience with Decision Trees, Logistic Regression, and Naïve Bayes, using 5-fold cross-validation to ensure robust performance.

4. Performance Metrics:

Learned evaluating models using metrics like accuracy, precision, recall, and F1-score, balancing interpretability and performance.

APPLICATIONS

- Revenue Management & Dynamic Pricing
- Customer Relations & Targeted Incentives
- Operational Efficiency & Workforce Planning
- Inventory Control & Resource Allocation
- Marketing & Branding Strategies
- Scenario Planning & Risk Management
- Continual Improvement via Feedback Loops

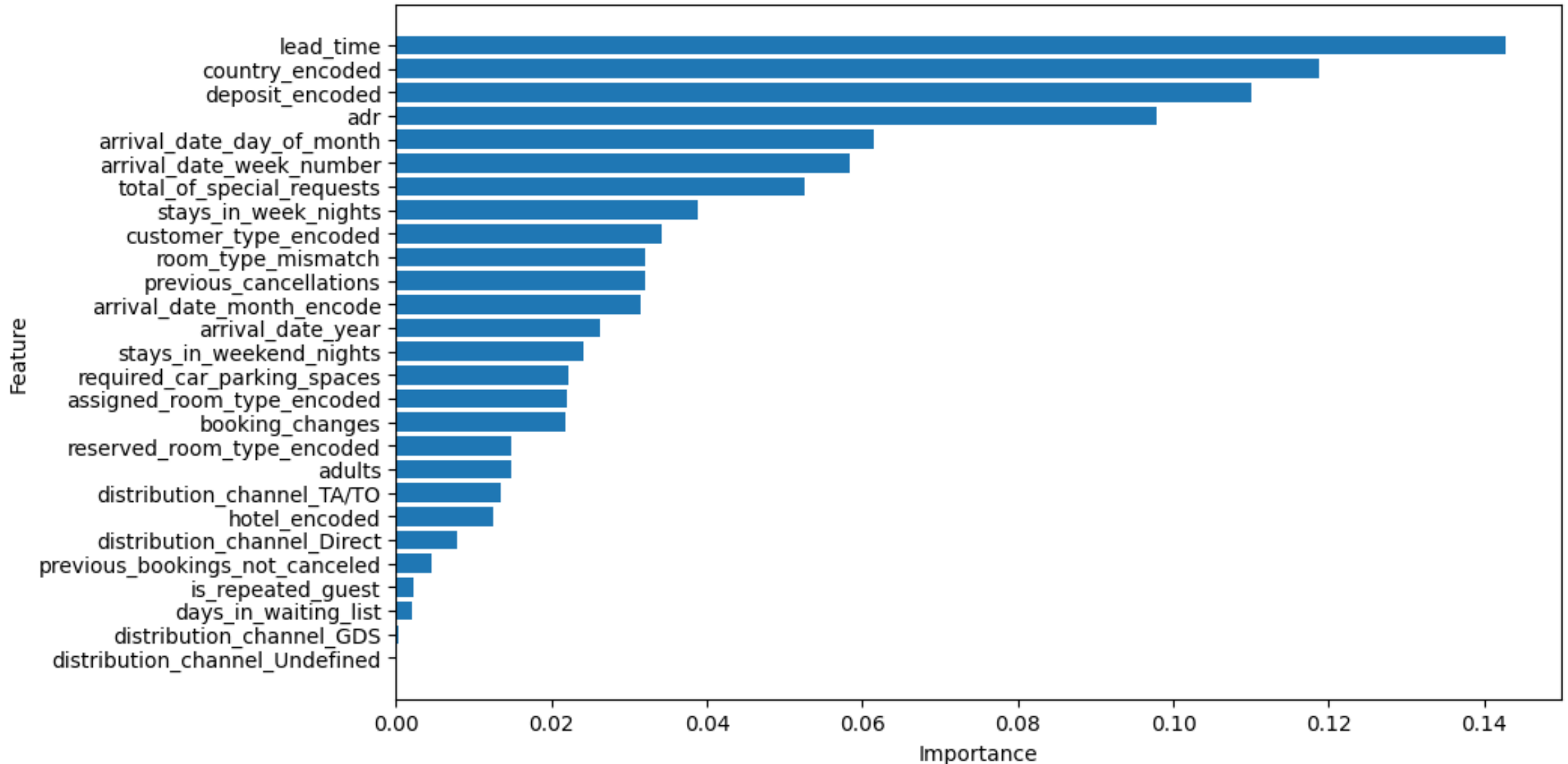
These applications leverage predictive insights into cancellation behavior to drive more informed, proactive decision-making.



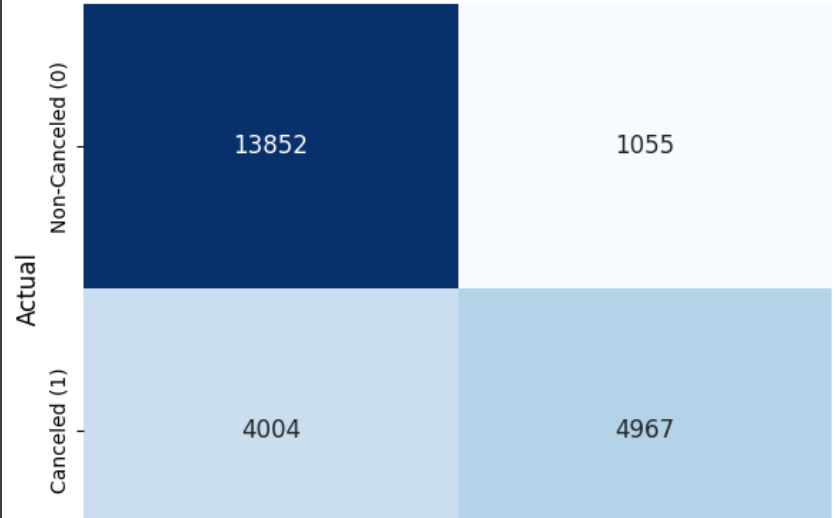
VISUALIZATION

Feature Analysis, Model Comparison, Confusion Matrix Heatmap, and ROC curve

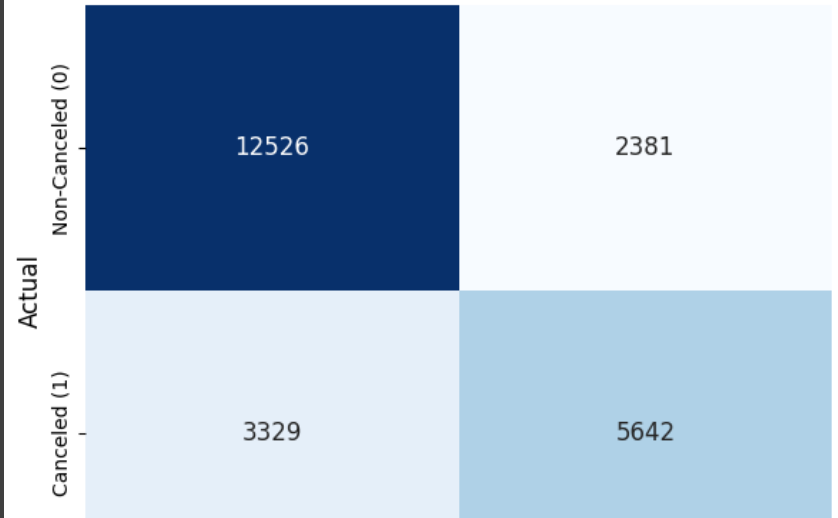
Feature Importance (Random Forest)



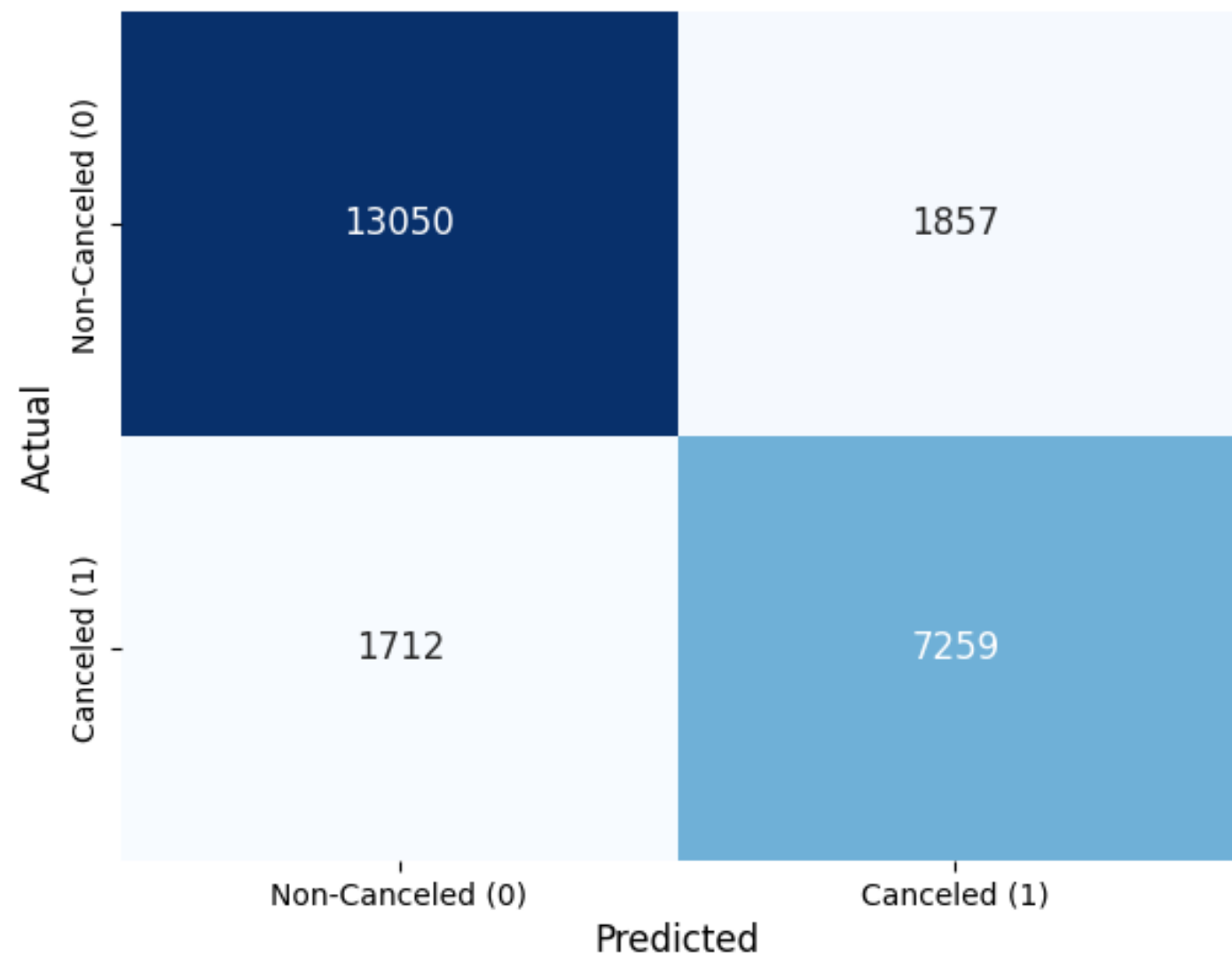
Logistic Regression Confusion Matrix

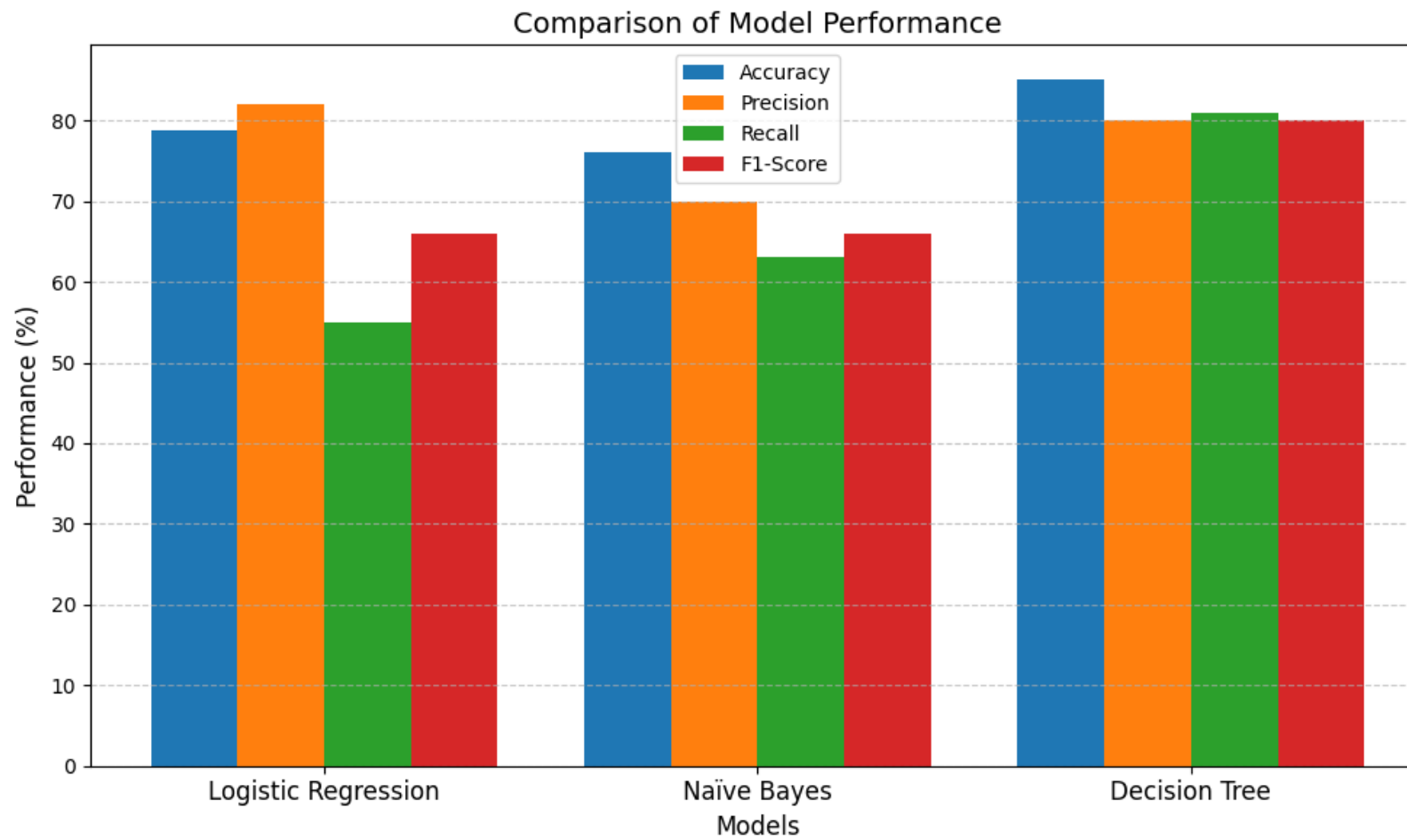


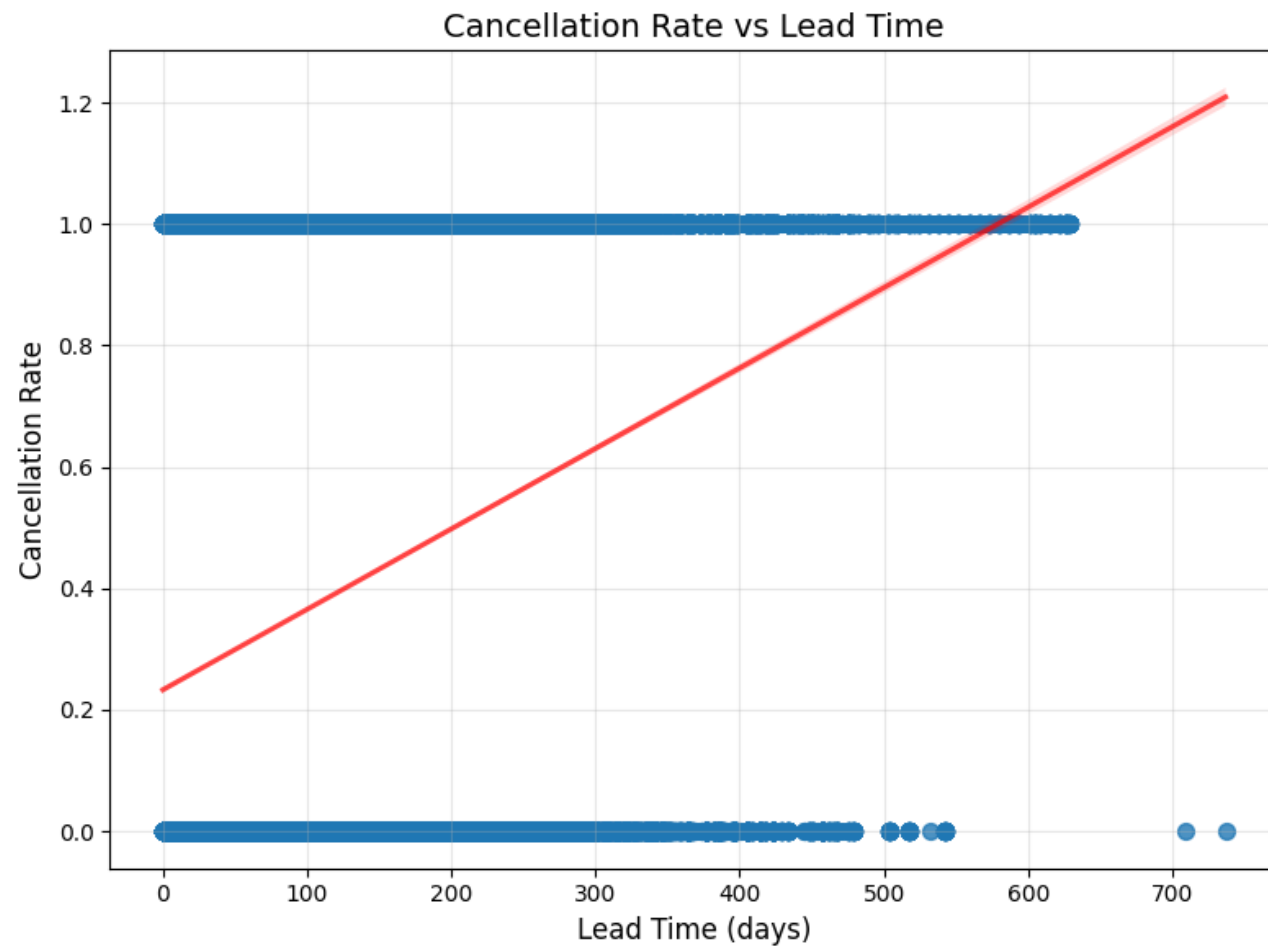
Naïve Bayes Confusion Matrix



Decision Tree Confusion Matrix







ROC Curves for Model Comparison

